「Def. Let X, Y be Metric Spaces, and $f_n : X \to Y$ be function defined for each $n \in \mathbb{N}$.

For each $x \in X$, define $f(x) \overset{\text{def}}{=} \lim_{n\to\infty} f_n(x)$ if the limit exists.

This f is called limit function of $\{f_n\}$.

「Def. Suppose that a sequence of functions $\{f_n\}$ on X into Y satisfies that;
the limit function f is defined on X and

For all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow d_Y(f_n(x), f(x)) \leq \varepsilon$, $\forall x \in E$.

Then, we say that $\{f_n\}$ converges uniformly on X to f. Denote $f_n \rightrightarrows f$.
Note: $\forall x \in X$, $d_Y(f_n(x), f(x)) \leq \varepsilon \Leftrightarrow \sup_{x\in X} d_Y(f_n(x), f(x)) \leq \varepsilon$. That is,

$$f_n \rightrightarrows f \text{ on } X \Leftrightarrow \lim_{n\to\infty} \sup_{x\in X} d_Y(f_n(x), f(x)) = 0.$$

「Thm. Let $\{f_n\}$ be functions defined on any Metric Space X into Complete Metric Space Y. Then,

$$\{f_n\} \text{ converges uniformly on } X \text{ to } f \ \underset{\text{In Complete Metric space}}{\overset{\text{In General Metric Space}}{\xRightarrow{\quad\quad}\!\!\!\!\xLeftarrow{\quad\quad}}}\ \forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } m, n \geq N \Rightarrow \sup_{x\in X} d_Y(f_n(x), f_m(x)) \leq \varepsilon$$

Proof. If $f_n \rightrightarrows f$, then,

$$\sup_{x\in X} d_Y(f_n(x), f_m(x)) \leq \sup_{x\in X} d_Y(f_n(x), f(x)) + \sup_{x\in X} d_Y(f(x), f_m(x)) \to 0$$

give the result. (간단해 보이지만, Supremum 의 성질과 triangle inequality 가 합쳐진 콤비네이션.)
Conversely, since Cauchy sequence in the Complete Metric Space converges, the limit function f exist.
Now,
$$\lim_{m\to\infty} \sup_{x\in X} d_Y(f_m(x), f_n(x)) = \sup_{x\in X} d_Y(f(x), f_n(x))$$

gives the result. (More specifically, let $n \in \mathbb{N}$ fixed. Observe that

$$\left| \sup_{x\in X} d_Y(f_m(x), f_n(x)) - \sup_{x\in X} d_Y(f(x), f_n(x)) \right|$$

$$\sup_{x\in X} d_Y(f(x), f_n(x)) = \sup_{x\in X} d_Y(\lim_{m\to\infty} f_m(x), f_n(x)) \overset{d_Y \text{ continuous}}{=} \sup_{x\in X} \lim_{m\to\infty} d_Y(f_m(x), f_n(x)) \overset{(*)}{=} \lim_{m\to\infty} \sup_{x\in X} d_Y(f_m(x), f_n(x))$$

$$\left| \sup_{x\in X} \lim_{m\to\infty} d_Y(f_m(x), f_n(x)) - \sup_{x\in X} d_Y(f_m(x), f_n(x)) \right| \leq \sup_{x\in X} \left| \lim_{m\to\infty} d_Y(f_m(x), f_n(x)) - d_Y(f_m(x), f_n(x)) \right| \to 0$$

gives (*).

Uniform Convergence and Continuity.

Thm. Suppose that $f_n, f: X \to Y$ are functions where X is Metric, Y is Complete Metric sp.
Let $x \in X$ be a limit point of X.
If for each $n \in \mathbb{N}$, the limit $\lim_{t \to x} f_n(t)$ exists and $f_n \rightrightarrows f$ on X,
then $\lim_{t \to x} \lim_{n \to \infty} f_n(t) = \lim_{n \to \infty} \lim_{t \to x} f_n(t)$

Proof: Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $m, n \geq N \Rightarrow d_Y(f_n(t), f_m(t)) \leq \varepsilon$, $\forall t \in X$.
Therefore, for this N and $m, n \geq N$, $\lim_{t \to x} d_Y(f_n(t), f_m(t)) = d_Y(\lim_{t \to x} f_n(t), \lim_{t \to x} f_m(t)) \leq \varepsilon$.

That is, $\{\lim_{t \to x} f_n(t)\}$ is a Cauchy sequence in Complete Y, it has a limit, say to A.

Meanwhile, since $f_n \rightrightarrows f$, we can choose $N_1 \in \mathbb{N}$ s.t. $n \geq N_1 \Rightarrow d_Y(f_n(t), f(t)) \leq \frac{\varepsilon}{3}$, $\forall t \in X$.
And, choose $N_2 \in \mathbb{N}$ s.t. $n \geq N_2 \Rightarrow |\lim_{t \to x} f_n(t) - A| \leq \frac{\varepsilon}{3}$.

And, since the limit exists, for this $n \geq N_2$, take $\delta > 0$ s.t.

$$t \in B_\delta(x) \setminus \{x\}, \quad |f(t) - \lim_{t \to x} f_n(t)| \leq \frac{\varepsilon}{3}$$

Finally, $t \in B_\delta(x) \setminus \{x\}$ and $n \geq \max\{N_1, N_2\}$,

$$|f(t) - A| \leq |f(t) - f_n(t)| + |f_n(t) - \lim_{t \to x} f_n(t)| + |\lim_{t \to x} f_n(t) - A| \leq \varepsilon.$$

☑

Corollary. If $\{f_n : X \to Y\}$ is a sequence of Continuous maps on a Metric space X into Complete Y, and $f_n \rightrightarrows f$, then f is Continuous.

「Uniform Convergence with Continuity

「Thm. Suppose that f_n $(n \in \mathbb{N})$ are functions on a Metric X into Complete Metric Y,
Let $x \in X$ be a limit point of X.
If for each $n \in \mathbb{N}$, the limit $\lim_{t \to x} f_n(t)$ exists and f_n converges uniformly on X,
then
$$\lim_{t \to x} \lim_{n \to \infty} f_n(t) = \lim_{n \to \infty} \lim_{t \to x} f_n(t).$$

Proof. By the Uniformly Convergence, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t
$$n, m \geq N \Rightarrow d_Y(f_m(t), f_n(t)) \leq \varepsilon \text{ for any } t \in X.$$

Uniform Convergence and Integration.

Thm. Let $\alpha : [a,b] \to \mathbb{R}$ be monotone increasing and $f_n \in \mathcal{R}(\alpha)$, $n \in \mathbb{N}$.
If $f_n \rightrightarrows f$ on $[a,b]$, then $f \in \mathcal{R}(\alpha)$ and

$$\lim_{n\to\infty} \int_a^b f_n \, d\alpha = \int_a^b \lim_{n\to\infty} f_n \, d\alpha = \int_a^b f \, d\alpha.$$

Proof. For each $n \in \mathbb{N}$, define $\varepsilon_n = \sup_{x\in[a,b]} |f_n(x) - f(x)|$. Then, clearly,

$$f_n - \varepsilon_n \leq f \leq f_n + \varepsilon_n \qquad (\text{for all } t \in [a,b])$$

Therefore, $$\int_a^b f_n - \varepsilon_n \, d\alpha \leq \underline{\int_a^b} f \, d\alpha \leq \overline{\int_a^b} f \, d\alpha \leq \int_a^b f_n + \varepsilon_n \, d\alpha \qquad (*)$$

Hence $$0 \leq \overline{\int_a^b} f \, d\alpha - \underline{\int_a^b} f \, d\alpha \leq 2\,\varepsilon_n \cdot [\alpha(b) - \alpha(a)] \to 0 \text{ as } n \to \infty.$$

Thus $f \in \mathcal{R}(\alpha)$. The equality is obtained by $(*)$, using $f \in \mathcal{R}(\alpha)$.

Uniform Convergence and Differentiation.

Thm. Let $f_n : [a,b] \to \mathbb{R}$ be differentiable maps s.t for some $x_0 \in [a,b]$, $\{f(x_0)\}$ converges.
If $\{f_n'\}$ converges uniformly on $[a,b]$, then $\{f_n\}$ converges uniformly on $[a,b]$ to f with

$$\frac{d}{dx} \lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \frac{d}{dx} f_n(x).$$

Proof. Given $\varepsilon > 0$, take $N \in \mathbb{N}$ s.t $n, m \geq N$ implies

$$|f_n(x_0) - f_m(x_0)| \leq \frac{\varepsilon}{2} \text{ and } |f_n'(t) - f_m'(t)| \leq \frac{\varepsilon}{2(b-a)}, \ \forall t \in [a,b].$$

By the Mean-Value Theorem for $f_n - f_m$, for all $x, t \in [a,b]$,

$$\left| f_n(x) - f_m(x) - \left(f_n(t) - f_m(t) \right) \right| \leq \frac{\varepsilon}{2(b-a)} \cdot (b-a) = \frac{\varepsilon}{2}. \qquad (*)$$

Therefore, for any $x, t \in [a,b]$ and $n, m \geq N$,

$$|f_n(x) - f_m(x)| \leq |f_n(x) - f_m(x) - f_n(x_0) + f_m(x_0)| + |f_n(x_0) - f_m(x_0)| \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

so that $\{f_n\}$ converges uniformly, let $f = \lim_{n\to\infty} f_n$. For each $x \in [a,b]$, define on $[a,b] \setminus \{x\}$

$$\phi_n(t) = \frac{f_n(t) - f_n(x)}{t - x} \quad \text{and} \quad \phi(t) = \frac{f(t) - f(x)}{t - x}$$

$(*)$ gives ϕ_n is Cauchy sequence, thus converges uniformly, to ϕ. Finally, the fact that ϕ_n continuous gives the result.

「Summary. Let $f_n : X \to Y$ be functions, $f_n \to f$.

Continuity: If f_n are continuous and $f_n \rightrightarrows f$, then f is continuous.

Differentiation: If f_n are differentiable and $f_n' \rightrightarrows f'$ and $f_n(x_0) \to f(x_0)$, then $f_n \rightrightarrows f$ and

$$\frac{d}{dx} \lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \frac{d}{dx} f_n(x)$$

Integral: If $f_n \in \mathcal{R}(\alpha)$ and $f_n \rightrightarrows f$, then $f \in \mathcal{R}(\alpha)$ and

$$\int_a^b \lim_{n\to\infty} f_n \, d\alpha = \lim_{n\to\infty} \int_a^b f_n \, d\alpha$$